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Attention graph convolutional nets for esophageal contraction pattern recognition in high-resolution manometries

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ABSTRACT

Diagnosis of the esophageal motility disorders is ongoing in clinical evaluations, which is based on traditional method called High-resolution manometry (HRM). However, the huge raw swallow data sets from the HRM are not allowed the doctors to interpret and classify the patients with esophageal symptoms. To this end, modeling propagation between vigor is useful for recognizing esophageal contraction patterns in large-scale high-resolution manometries. In this paper, we learned a discriminative propagation between vigor using deep learning methods. Furthermore, we designed an efficient graph to incorporate contractile vigor propagation (CVP) that considers the contraction and pressure propagation between vigor in HRM images. Using an attention graph convolutional network (GAT), the edges in CVP can automatically learn the contraction patterns (i.e., local temporal information) trends in time series and propagation (i.e., global information) through the attention units. The attention mechanism layer leverages the short-term trend to improve the prediction accuracy. The quantitative experiments showed that the proposed method achieves high accuracy in esophageal contraction pattern recognition and demonstrates its effectiveness compared to existing traditional methods. We also visualized the learned vigor-specific propagation patterns in the contractile features, which show that the proposed CVP-GAT is able to develop interpretable propagation information for esophageal contraction pattern recognition.

1. Introduction

High-revolution manometry (HRM) is a gold standard method/technique used in the clinical routine to diagnose the esophageal motility disorders [1] and was first introduced since 1950s. When patients with esophageal symptoms (chest pain, heartburn and regurgitation) do not respond to acid suppression therapy and their symptoms are not explained by gastroscopy, we should consider that they have esophageal motor disorders [2,3]. In clinical evaluation, measuring the pressure of the lower esophageal sphincter (LES), the water-perfused or solid-state was used and it was tremendous scare to record. HRM incorporates up to 36 circumferential sensors, and the data is converted into seamless and dynamic spatiotemporal esophageal pressure topography (EPT) by advanced software algorithms [4]. This gold standard, however, has received potential attention in recent years due to its revolution to automatically provide useful information for the functionality of esophageal motor, in which new metrics and parameters were incorporated to facilitate its interpretation and classification results [5,6]. Furthermore, despite the information provides by HRM during swallow, the doctors cannot fully analyze and interpret the pressure recoded at each time. Hence, to cope with this problem Chicago Classification has been proposed to analyse and interpret the measuring pressure recording based on different thresholds [7–10]. The architecture of the Chicago Classification comprised four important parts: incomplete LES, major motility disorders, minor motility disorders, and normal esophageal motility [11]. Each part has its own function in the system. These method has their own pros and cons.

Although the introduction of the Chicago Classification has facilitated to analyse and quantify the effect of bridge position swallow on esophageal motility individuals [12]. In their study the authors found

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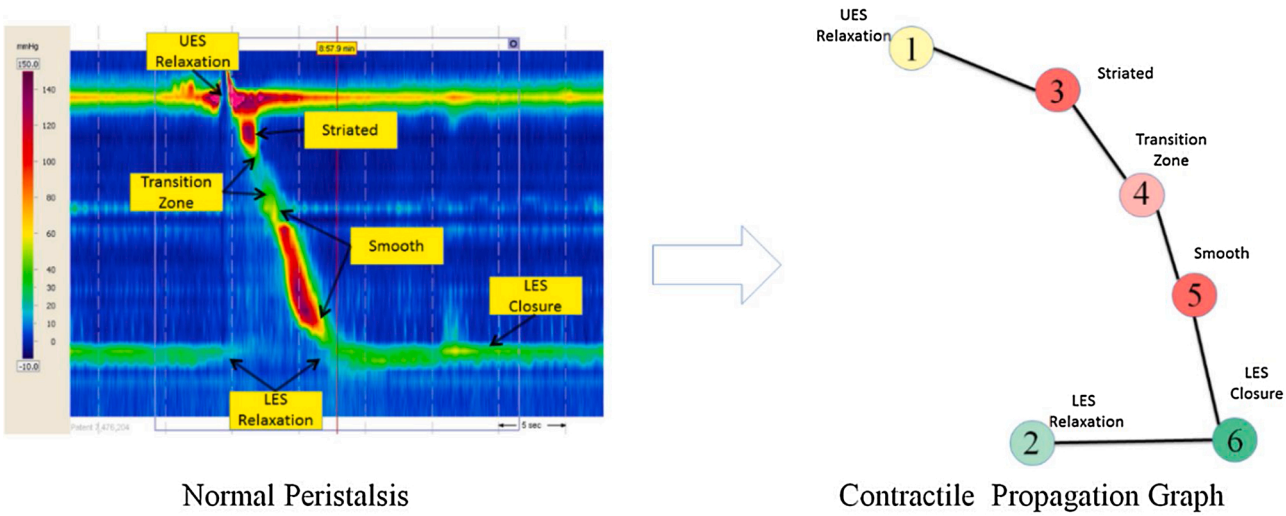

Fig. 1. Understanding the contraction process from an individual swallowing scene requires accurately identifying the relevant contractile propagation between vigor.

that among the measuring elements pressure including the proximal contractile integral and distal contractile integral, integrated relaxation pressure, distal latency, peristaltic beak, intrabolus pressure, and expiratory and inspiratory esophageal junction considered in the bridge position, only the intrabolus pressure was statistically significant higher than spine. Another recent study also used HRM based on the Chicago classification to predict the effect of the peroral endoscopy myotomy (POEM) for the treatment of the achalasia [13]. After the performance of the HRM on the POEM the authors observed that there is no prediction of the outcome, but remarkable significantly change was observed in the integrated relaxation pressure and contractile pattern. This may be due to the sensitivity of the Chicago Classification to noise that may be led to a high error and false analysis of the results. The main limitation of the above mentioned studies is that the result obtained from Chicago classification cannot allow the doctors to get the false positive and false negation result, i.e., the patients who have essophageal symptoms are classified as negative and those who have not essophageal symptoms are classified as positive. Moreover, the esophageal motility disorders are not often explained by HRM system, which constituted the major drawbacks of the HRM [14].

To date, these drawbacks are crucial challenges tacks in clinical practice of the esophageal motility disorders during the HMR studies. To address this issue, a novel method such as deep leaning method can be adopted to assist the healtcare in clinical evaluation of the esophageal symptoms. However, this deep learning method by using the esophageal contraction pattern recognition from high-resolution manometry, which plays a crucial role in diagnosing specific clinical disorders [15,16] can be used to classify the patients with esophageal symptoms in the few minute and low error. More recently, to classify and reduce features extracted bias from the HRM images, variational autoencoder was proposed to assist the doctors during the diagnosis of the esophageal motility disorders [17]. The authors method where based on dimension reduction to find the useful features (contraction patterns) from HRM images. The authors found that the latent variables considered in four dimensional, only the strength, propagation pattern and sphincter relaxation were encoded with low error on training and testing dataset. In addition, among the loss considered in their studies only the hybrid L2 with regularization parameter was able to select the important features during the contraction and relaxation transition. In similar fashion, to identify the spastic (type III) and non-spastic (type I and II) of achalasia subtype, supervised machine learning method such as decision tree and random forest algorithm were used on the HRM images [18]. A total of 180 HRM data was used to predict the achalasia subtypes, the authors found that to classify spastic (type III) and non-spastic (type I and II) used train data was 90 % in accuracy, while for testing data was 78 % in accuracy. In addition, the accuracy of achalasia subtypes I, II and III used train and test data were 71 % and 55 %, respectively. The limitation of the above mentioned methods is that the HRM data are often imbalance which can lead to bias in the accuracy. The authors did not take into account the correlation that exist among the features.

Furthermore, to comprehend an esophageal contraction vigor scene, the model could not only identify the pressure position of the contextual information but also infer the temporal activity. The esophageal contraction pattern can be evaluated through HRM images, which illustrate a combination of time, amplitude and length of the peristaltic wave. The distal contractile integral (DCI) is used to measure the vigor of the contraction. Moreover, the amplitude (mmHg), duration (seconds), and length of the contraction between the proximal and distal regions produce the DCI value. DCI identifies peristalsis waves as failed, weak or normal. The ability to accurately capture the relevant association with pressure and utilize relational reasoning is important for comprehending the vigor process of esophageal contraction [19–24]. However, modeling the propagation between vigor is a challenging task, because the above mentioned studies have access only to pressure labels and did not have any knowledge of the underlying association information. Therefore, it is necessary to infer the propagation between vigor from other aspects when designing effective deep methods for understanding esophageal contraction vigor.

To capture the contraction and propagation between vigor to achieve esophageal contraction pattern recognition, we aim to model vigor propagation through a more efficient and flexible approach, by learning the graph-structure data between vigor automatically from HRM images; then, inference for esophageal contraction pattern recognition could be performed efficiently. Specifically, we propose modeling contractile vigor propagation by building a contractile vigor propagation graph (CVP), as illustrated in Fig. 1, in which the graph nodes represent contractile features, and the edges represent the correlation between two vigor occurrences. In our experiments, we applied a state-of-the-art two-dimensional convolutional neural network (ConvNet) to form a unified contraction event recognition framework. Based on the graph convolution output [25], the CVP edges can be automatically optimized in an end-to-end fashion. Our proposed model not only recognizes contraction and propagation in an individual swallowing scene but also instantaneously generates a contraction-specific vigor propagation graph to facilitate further insights into esophageal contraction vigor.

Our network learns to represent the scene via a vigor propagation

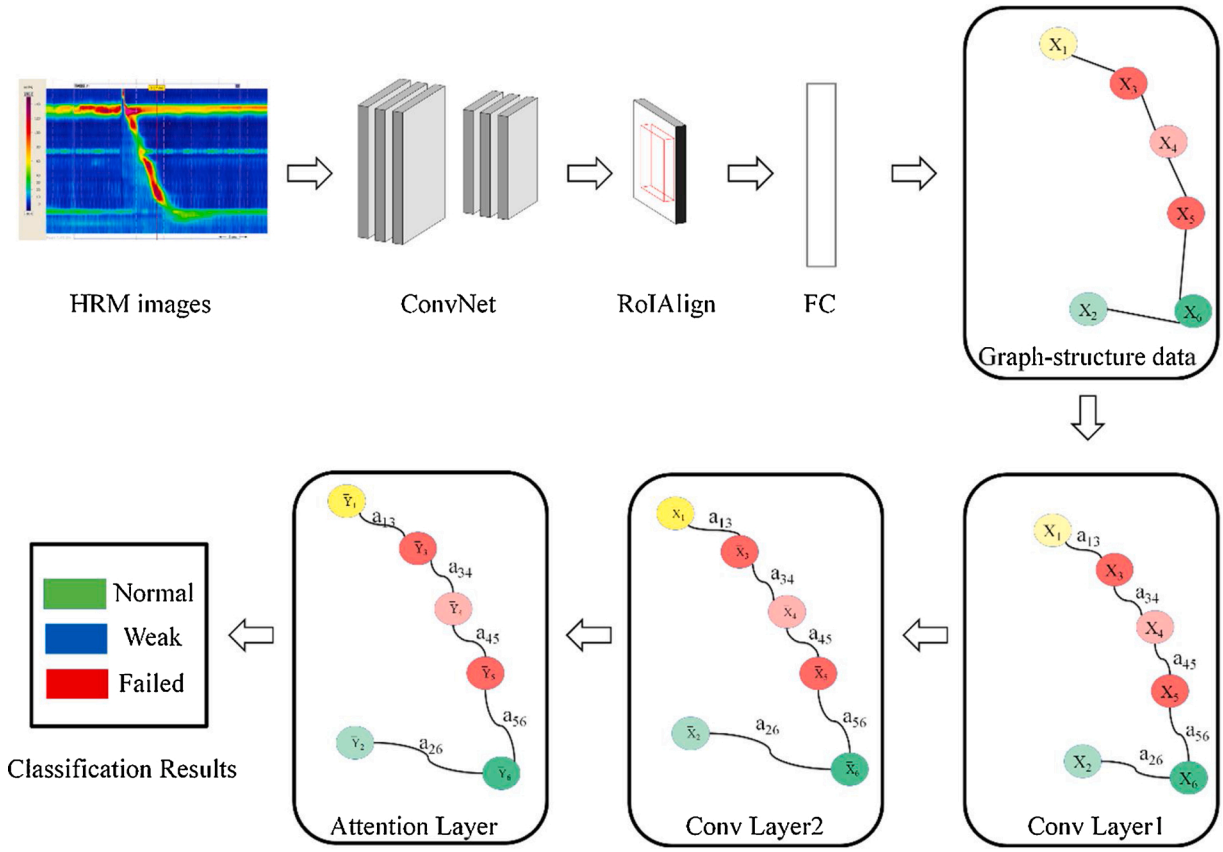

Fig. 2. An overview of our model framework for esophageal contraction pattern recognition.

graph and then uses that to perform reasoning about the contraction process ('Normal Peristalsis' in the showed example) based on the graph structure and node features. The normal peristalsis comprises six parts: The upper esophageal sphincter relaxation (UES), striated, transition zone, smooth, the lower esophageal sphincter relaxation (LES) and the lower esophageal sphincter closure. An vigor is denoted as a node (from 1 to 6), and the propagation between contractions are illustrated as edge.

In order to improve the modeling efficiency of vigor propagation HRMs, the optimized node and sparse matrix techniques are used in the CVP. Specifically, for spatial feature, we observe that contractile propagation is naturally contraction prior, where UES, Resting and LES are densely captured but the pressure changes are subtle. The upper esophageal sphincter relaxation (UES) plays a crucial role during the swallowing and regulates the pressure to allow the bolus to pass [26]. The striated has been as an important valuable of esophageal motor disorders during the swallowing. Moreover, the mean amplitude of contractions in the distal esophagus and the duration of contractions for the length of the distal esophageal segmentation represents the DCI value which was transferred to the proximal aspect of the lower esophageal sphincter relaxation (LES) and the lower esophageal sphincter closure through the transition zone and smooth [12] (Fig. 1). Thus, rather than connecting pressure patterns, we create a sparse CVP by dropping several pressure activities and forcing connections only between vigor within a local neighborhood.

Recently, the graph convolutional network (GCN) has become a semi-supervised model that can process graph-structured data. GCNs have achieved promising results in many applications, such as image classification [27], document classification [28], and unsupervised learning [25]. GCNs use two convolutional modes, namely, spectrum approaches [29] and spatial domain approaches [28]. Petar et al. [30]. proposed the graph attention network (GAT) by using a self-attention mechanism in a GCN model. In deep learning, attention mechanisms have been widely used to focus network attention on the most relevant input features. The specific implementation of GAT contributed the attention, which computes coefficients between adjacent nodes and indicates how close those two nodes are in the feature space.

In this work, we introduce a novel CVP graph attention network (CVP-GAT) to recognize esophageal contraction patterns from high-resolution manometry images. Our study explicitly model the contractile features to build flexible and explainable CVP, that differs from the peristalsis. Furthermore, we adopt attention algorithm with a sparse propagative sampling strategy, that enables efficient contractile propagation inference in an efficient manner. The major contributions of our proposed algorithm consist of the following aspects:

- We designed efficient and flexible vigor propagation graphs while simultaneously capturing the contraction and pressure propagation between vigor to enable esophageal contraction pattern recognition. We provide an interpretable mechanism to explicitly model the relevant propagation among vigor in a swallowing scene and thus ability to discriminate between different esophageal contraction vigor occurrences.
- We propose an efficient captioning strategy over the vigor propagation graphs by utilizing GAT with a sparse spatial modeling scheme. The introduced network has the potential to conduct contractile propagation reasoning over contractile propagation for esophageal contraction pattern recognition.
- The proposed network yields the state-of-the-art performances on HRM images. The visualizations of the learned vigor propagation graphs and contractile features demonstrate that our approach can propagate vigor information for use in esophageal contraction pattern recognition.

2. Methodology

Our goal is to recognize the esophageal contraction pattern in an individual swallowing scene by explicitly exploiting vigor propagation features. Therefore, we design contractile vigor propagation (CVP) to represent an individual swallowing process, and employ propagated captioning on it for esophageal contraction pattern recognition. In this section, we provide detailed descriptions of our method. First, we present an overview of our model. Then, we explain how the CVP is built. Finally, we introduce the efficient training and inference approaches for CVP.

2.1. Framework of esophageal contraction pattern recognition

The framework of the proposed scheme is illustrated in Fig. 2. Given an individual HRM image as input and the bounding boxes of the swallowing areas in the scene, our scheme takes three crucial steps. First, a set of contractile vigor are uniformly sampled from the high-resolution manometry, then feature vectors of vigor propagation are extracted from sampled contractile vigor by applying ConvNet. The output of the ConvNet represent the features extraction of the vigor propagation. Hence, inspired by the object instance segmentation strategy [31], which efficiently detects features in an image and simultaneously generates a high-quality segmentation label for each instance. Furthermore, it is worth to mentioned that the Mask R-CNN is an extension of the Fast R-CNN [32] and also simple representative of the object segmentation algorithm. Its principal function is to avoid misalignment of the prediction object detection keypoints in the images by preserving the exact spacial location of the objects (called RoIAlign which is the main part that we explore in our experiment to quantify each feature extracted from the HRM images using ConvNet (Fig. 2), and consists of the two important parts: (1) the foreground and background are used to make the difference between the instance region based on the region proposal network (RPN), (2) the fully connected network was applied to classify and segment the object region. Moreover, we also applied RoIAlign [31] to detect the bounding box of each vigor instance from the contractile vigor map. Then, a fully connected (FC) layer is employed on the aligned features to obtain a representation feature vector for each contractile vigor.

We first extract feature vectors of contractile vigor from high-resolution manometry images. Contractile vigor propagation graphs are built to capture the information propagated among contractile vigor. Next, an attention graph convolutional network is used to perform propagated reasoning on graphs. All the output graphs are then fused to produce the propagated feature vectors of contractile vigor. Finally, the original feature and the propagated feature are aggregated and input into contractile process classifiers from an individual swallowing scene.

After provided these representation features of contractile vigor, we build vigor propagation graphs, where each vigor is depicted as a node and their distance of two vigor representation features is computed as a scalar weight of each connection in the graphs. Finally, we utilize learning of recognize vigor propagation. We execute the GAT to build correlational reasoning based on CVP. Graph convolution fuse the CVPs together to generate correlational representations for vigor. Then, a classifier for recognizing vigor propagation is applied to the pooled vigor correlational representation and a fully connected layer is applied on the HRM image for vigor instance classification. Upon maxpooled vigor representations, contraction-level representations are generated to classify contractile pattern by using another fully connected layer. Lastly, the activation function softmax was used to classify patients with esophageal motor disorders in three classes including normal, weak, and failed (Fig. 2).

2.2. Building contractile vigor propagation graphs

Contractile vigor propagation (CVP) is the key component in our framework. To understand the esophageal contraction pattern, we utilize a graph structure information to explicitly model vigor propagation. Our efforts were inspired by relational reasoning [33] and graph neural networks [25].

2.2.1. Graph definition

Typically, the nodes in contractile vigor propagation graph expressed as a vigor scheme $V = \{(a_i,\ c_i) \mid i = 1,\cdots, N\}$, where N is the number of vigor, $a_i \in R^d$ is the feature of vigor i^{th} feature, and $c_i = (x_i, y_i)$ is the center coordinate of the bounding box of vigor i. Graph G $\in R^{N\times N}$ represents propagation among vigor, where correlation value G_{ij} encodes the jth importance feature to vigor i.

In order to improve representational power, we considered both representation feature and position information to obtain the underlying correlation between two vigor. Notably, representation correlation and position correlation have various semantic characters. Finally, our model learns the representation correlation and position correlation in explicit and a separate approach. A composite function of the correlation value can be calculated as follows:

$$G_{ij} = f\left(f_a\left(a_i, a_j\right),\ f_c\left(c_i, c_j\right)\right) = \frac{f_c\left(c_i, c_j\right) exp\left(f_a\left(a_i, a_j\right)\right)}{\sum_{i=1}^{n} f_c\left(c_i, c_j\right) exp\left(f_a\left(a_i, a_j\right)\right)} \tag{1}$$

where $f_a(a_i, a_j)$ denotes the representation correlation between two vigor, and the position correlation is computed by $f_c(c_i, c_j)$. The function f fuses the representation and position correlation to a scalar weight. We perform normalization on each vigor node to ensure that the sum of all the correlation values of one vigor node i will be 1.

2.2.2. Representation correlation

Here, we discuss the computing representation correlation value between contractions. In the spatial dimension, the representation features can be affect each other, because the information from each node may not be the same. Then, to cope with this issue, the spatial representation correlation among the contractions can be adjusted. From the definition of the dynamic spatial correlation in [34], we can calculate the spatial representation correlation as follows:

$$f_a\left(a_i, a_j\right) = ReLU\left(W_a\left(a_i, a_j\right) + b_a\right) \tag{2}$$

where $(\cdot,\cdot)$ concatenate i^{th} and j^{th} representation features. W_a and b_a are the reward learned weights and bias. The concatenated vector is transformed to a scalar activated by a *ReLU* nonlinearity.

2.2.3. Position correlation

The position correlation is also used to add spatial structural information to the contractile graph. We use the recent approach in [35] to perform learning position correlation. Hence, from the definition of the dynamic temporal correlation in [34], we can also calculate the position correlation as follows:

$$f_c\left(c_i, c_j\right) = ReLU\left(W_c \Phi\left(c_i, c_j\right) + b_c\right) \tag{3}$$

Φ is a simple embedding high-dimensional feature and computed the relevant distance between two vigor, such as cosine and sine functions of different lengths of contractile vigor. Let dc be the representation dimension after embedding. Then, we transform the embedded representation into scalar weight vectors (W_c and b_c) activated by a *ReLU* function.

2.3. Reasoning and training on GAT

After the CVPs are constructed, we can perform correlational reasoning on them for pattern recognition of propagation vigor. We start by describing a single graph attentional layer, which is utilized

throughout all the GAT architectures in our experiments. The attention mechanism is based on the encoder-decoder model, which has been applied in neural machine translation tasks [36]. Currently, attention mechanisms are widely used in recommendation systems [37], image caption generation [38] and document classification [39]. From a development perspective, attention mechanisms can be categorized into multiple types, such as self-attention [35], global and local attention [40], and soft and hard attention [36]. In this study, we used a soft attention model to learn the important information, and then calculated a context vector to express the global variation trends of vigor propagating tasks.

Suppose that a contraction series is represented by x_i $(i = 1,2,\cdots, n)$, where n is the contraction series length. The design process of the soft attention mechanism is as follows. First, the hidden states h_i $(i = 1,2,\cdots, n)$ are calculated using instance segmentation (and their variants); these are expressed as $H = \{h_1, h_3,\cdots, h_n\}$. Second, a scoring function is designed using a multilayer perceptron to calculate the score for each hidden state. Third, an attention function is employed to compute the context vector (Cont) that describes the global contractile variation information. Finally, the output results are calculated using the context vector.

In particular, we used the characteristics hi as input when computing the weight of each hidden state. The corresponding outputs are obtained through two hidden layers. The weights for each characteristic (α_i) are computed by a *softmax* normalized index function (Eq. 4), where w1and b1are the weight and deviation of the first layer, respectively, and w2and b2are the weight and deviation of the second layer, respectively.

$$z_i = w_2(w_1 H + b_1) + b_2 \tag{4}$$

$$a_i = \frac{exp(z_i)}{\sum_{k=1}^{n} exp(z_k)} \tag{5}$$

Finally, we designed the attention mechanism. Furthermore, this attention mechanism was applied to focus on the important output contractile information (maximum information) from two hidden layers, which constituted of the sequence context vector (Cont), called scalar dot-product attention [37] and covers the global contractile variation information. The formula is defined in Eq. 6 as follows.

$$Cont = \sum_{i=1}^{n} a_i * h_i \tag{6}$$

The entire model is trained using an end-to-end approach with backpropagation. Combined with standard cross entropy loss, the final loss function is

$$L = L_1\left(y^C, \hat{y}^C\right) + \lambda L_2\left(y^P, \hat{y}^P\right) \tag{7}$$

where L_1 and L_2 are the cross-entropy loss, y^C and y^P denote the ground-truth labels of contractile vigor and vigor propagation, and $\hat{y}^C$ and $\hat{y}^P$ are the predictions of contractile vigor and vigor propagation. The first term corresponds to the contractile vigor classification loss, and the second is the loss of the vigor propagation classification. The weight λ is used to balance these two tasks.

In summary, we propose the CVP-GAT to achieve vigor propagation classification of esophageal contraction. The high-resolution manometry image is constructed into a graph network, where the contractile state is described as node attributes. The topological characteristics of the contraction are captured by an instance segmentation network to obtain spatial dependence. The dynamic variation in node attributes is captured by a GCN to obtain the local temporal tendency of the contractile state. The global variation trend of the contractile state is then captured by the attention model, which is conducive for realizing accurate contractile vigor propagation.

3. Experiments

In this section, we first introduce the experimental dataset and the metrics used to evaluate our method. Then, we describe a number of experimental studies performed to understand the contributions of the proposed components in our model. We also compare the performance of our approach with that of other state-of-the-art models. Finally, we visualize our learned contractile vigor propagation relation graphs and features.

3.1. Data description

High-resolution manometry images were obtained by gastroenterologists using a recording system (Manoscan™, Sierra Scientific instruments) and a dedicated computerized system (Mano View, Given Imaging, Los Angeles, CA, USA). The study was approved by the Ethics Committee of Xiangya Hospital, Central South University. Informed consent regarding the publication of this article was obtained from the subjects. The dataset is composed of 4000 high-resolution manometry (HRM) images gathered from 400 individual patients. To classify and identify each patient with esophageal motility symptoms correctly and reduced the misclassification error, we labeled our dataset by using the threshold used in clinical practice base on the DCI values [41]. Hence, the threshold of the DCI values into failed (DCI < 100 mmHg.cm.s), weak (100 mmHg.cm.s < DCI < 450 mmHg.cm.s), and normal (DCI > 450 mmHg.cm.s) generated 1450 images of failed peristalsis, 1050 images of weak peristalsis and 1050 images of normal peristalsis, respectively. After labeled the dataset into three classes, we then split the dataset respectively into train (70 %), validation (10 %), and test (20 %) to experiment our proposed model performance.

3.2. Evaluation metrics

We evaluated the prediction performance of the model using the error between the real vigor propagation and the predicted results based on the following metrics:

(1) Precision (Pre):

$$Pre = \frac{y_i^{tp}}{y_i^{tp} + y_i^{fp}} \tag{8}$$

(2) Recall (Rec):

$$Rec = \frac{y_i^{tp}}{y_i^{tp} + y_i^{fn}} \tag{9}$$

(3) F1-score (F1):

$$F1 = 2 \times \frac{Pre \times Rec}{Pre + Rec} \tag{10}$$

(4) Coefficient of Determination (R2):

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} \left(y_i - \hat{Y}_i\right)^2} \tag{11}$$

(5) Accuracy (Acc):

$$Acc = 1 - \frac{\| Y - \hat{Y} \|_F}{\| Y \|_F} \tag{12}$$

Table 1
Ablation studies for contractile propagation vigor pattern recognition.

a. Exploration of representation correlation.	
Method	Accuracy
base model	88.6 %
concatenation	95.3 %

b. Exploration of position correlation.	
Method	Accuracy
no position correlation	90.7 %
distance encoding	95.5 %

c. Exploration of number of nodes.					
Number	10	50	100	200	500
Accuracy	87.8 %	95.8 %	91.1 %	90.9 %	89.1 %

d. Exploration of attentional modeling method.	
Method	Accuracy
no attention	92.1 %
attention layer	96.3 %

here, we consider for each class y_i^{tp}, y_i^{fp}, y_i^{fn} and y_i^{tn} as true positives (mean that the patient is positive and he/she is classified as positive), false positive (mean that the patient is negative and he/she is classified as positive), false negatives (mean that the patient is positive and he/she is classified as negative), and true negative (mean that the patient is negative and he/she is classified as negative) [42,43], respectively, on HRM i, and y_i and $\hat{y}_i$ are the real and predicted vigor propagation on HRM i, respectively. N is the number of nodes on HRM. Y and $\hat{Y}$ are the sets of y_i and $\hat{y}_i$, respectively.

Precision, Recall and the F1-score are used to measure the prediction error, while accuracy is used to measure the prediction of vigor propagation. A high accuracy value is preferred. R^2 represents the correlation coefficient, which measures the relative ability of the prediction results: a larger R^2 value reflects a better prediction effect.

3.3. Experimental result analysis

Each vigor propagation is extracted to a 64-dimensional feature vector, using the methods mentioned in Section 2.1. Our backbone network utilize Mask-RCNN [31] during Experiments. We also experimented with a VGG [44] network to ensure a fair comparison with prior methods. The base model performs esophageal contraction instance segmentation on the original vigor features without performing correlational reasoning. Then, we further generate relational connection to complete CVP, that feed into GAT and train to learn vigor propagation.

For the training, we utilize ADAM with a learning rate ranging from 0.0001 to 0.001 to learn the model parameters and set hyper parameters to $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 10^{-8}$. The train epoch are 800 with Early Stopping and a mini batch size of 32. In addition, the loss weight $\lambda = 1$ is employed. The proposed network was trained and executed on a Dell XPS 8930 server equipped with a 6-core 3.20 GHz processor, 16 Gb RAM and one NVIDIA GeForce GTX 1070 video card. The proposed method was built on the Keras framework and runs on a TensorFlow back end.

Furthermore, we perform several ablation studies of the proposed model components to understand the contributions of correlation modeling for esophageal contraction pattern recognition and employ accuracy as an evaluation metric. The result are shown in Table 1.

3.3.1. Representation correlation

We begin our experiments by investigating the effect of learning the representation correlation between contractions and different functions to compute the representation correlation value. Based on a single HRM image, we first constructed a single CVP without using position correlation. The results are listed in Table 1-a, which shows that explicitly modeling the correlation between contractions provides a significant performance improvement, It yields a recognition accuracy of 95.3 % and outperforms other state-of-the-art networks. We assume that this concatenation operation is more reliable for representing correlation feature. In the successive experiments, we use embedded this concatenation to compute the representation correlation value.

3.3.2. Position correlation

Spatial structural information are added into CVP. In Section 2.2, to use spatial features, we perform the distance encoding method. The performance results of comparison of these methods are listed in Table 1-b. We can see that the distance encoding method obtains performance better than those without using spatial features and boosts slightly better accuracy 95.5 %. It indicates that modeling of the position relation is effective. In the rest of the paper, the position relation are represented by using distance mask.

3.3.3. Graphs

We further extend to explain the effectiveness of graph nodes, which capture different kinds of correlation information. First, the performance of different numbers of nodes are compared. As depicted in Table 1-c, nodes Ng = 50 leads to consistent and symbolic result in comparing to other nodes numbers, and further improve accuracy by 95.8 %. This observation demonstrates that different nodes numbers learn different semantic information, and more nodes will reduce recognition performance, i.e., correlational reasoning.

3.3.4. Graph attentional layer

We investigate the proposed model in the attention domain. As described in Section 2.3, a soft attention strategy [36] is utilized, and uniformly sample a set of $n = 6$ contraction series from the CVP in the training step. In the simplest configuration, the input contraction series are handled separately, and then calculate the context vector to cover global contractile variation information. Alternatively, we can build a propagational graph upon the contractile vigor in the input contraction series and cover global contraction information by a graph attentional layer. The accuracy results of contraction series modeling are reported in Table 1-d and it is helpful to for improving the recognizing performance of our model. Furthermore, the attentional algorithm further boosts the accuracy to 96.3 %, which demonstrates that attentional learning benefit to differentiate the propagation vigor of esophageal contraction.

Table 2
Comparison with state-of-the-art.

Method	Backbone	Case	Metrics				
			Acc(%)	Pre(%)	Rec(%)	F1(%)	R^2(%)
Szegedy *et al.* [45]	MASK-RCNN + Inception-v3	3	91.5 %	91.9 %	91.2 %	91.5 %	91.2 %
Karen *et al.* [46]	MASK-RCNN + VGG16	3	90.1%	90.4 %	89.8 %	90.9 %	89.8 %
Huang *et al.* [46]	CNN	2	95.6%	×	×	×	×
CVP-GAT	MASK-RCNN + GAT	3	95.8 %	96.5 %	97.5 %	95.2 %	93.5 %

Acc: accuracy; Pre: precision; Rec: recall; F1: F1-score; R2: coefficient of determination.

Table 3
Comparison between our proposed model, PANet, and PolarMask, measured by accuracy.

Method	Accuracy
PANet [48] + GAT	93.4 %
PolarMask [49] + GAT	94.1 %
CVP-GAT	95.8 %

3.4. Performance comparison

Here, we compare our model with other state-of-the-art approaches. To ensure a fair comparison with the prior approaches, we report the results of our method when using both Inception-v3 [45] and VGG [46] baseline networks. Huang el at [47]. trained a CNN to distinguish normal from abnormal swallowing on HRM data. Table 2 shows the comparison results for esophageal contraction pattern recognition.

From the result depicted in Table 2 we observed that our proposed method surpassed the existing methods ConNet of 0.2 %. This may be due to the significantly small size of the HRM dataset, which is scars challenges tasks in the clinical routine and constituted the limitation of our proposed model to achieves tremendous accuracy. In addition, we can achieve efficient and best accuracy compared to existing methods based on ConvNet when more raw HRM image dataset are available. Hence, exploring the HRM dataset available we concluded that our CVP-GAT model again achieves the state-of-the-art performance.

Meanwhile, we also perform a swallowing box specific experiment based on Mask-RCNN. Using the bounding box proposals from Mask-RCNN for inference, our model can still obtain higher recognition accuracies. Remarkably, our method outperforms a good margin when compared to the existing approaches and establishes a novel state-of-the-art method. However, it is worth noting that our method utilizes the instance segmentation strategy, and capable of better recognizing these challenging patterns, due to our proposed components can capture and exploit the correlation information among vigor.

In addition to a Mask-RCNN, identifying bounding boxes for proposal features is another way to enable information dissemination between proposals. Thus, we conducted another baseline test using ConvNet models (e.g., PANet [48], PolarMask [49]) on the proposal features and then created a bounding box on the PANet output over adjacent proposals. The adjacent connections are formulated by using the same graph as the GAT. The results and a comparison are listed in Table 3, i.e., our proposed method achieves on test data 95.8 % in accuracy, whereas those proposed in literatures in [48], PolarMask [49] were 93.4 % and 94.1 % in accuracy, respectively. This may be due to the reliability of the graph convolution network to automatically recognize the features in the heterogeneous data sets. Its outstanding performance demonstrates the effectiveness and generalizability of the proposed CVP for capturing the correlation information from esophageal contraction scenes. Moreover, the experiment conducted on the HRM image to classify the esophageal motility disorders using the metrics such as train loss, validation loss, and testing loss on our proposed model were $0.4184e^{-1}$, $0.44882e^{-1}$, and $0.42719e^{-1}$, respectively, while the train loss, validation loss, and test loss of the proposed methods in [17] were $3.772e^{-1}$, $3.787e^{-1}$, and $3.766e^{-1}$, respectively. From the above results, our method slightly outperforms that proposed in [17]. This may be due to the sample imbalance of the raw HRM dataset. We can achieve tremendous result, when more HRM data are available.

In Fig. 3, we visualize several examples of the correlation graph, which are generated by the proposed module. We use the graph generating method on a single HRM image, because it is easier to visualize. Illustration of visualization facilitate us to understand that, how CVP works. We can observe that our model can gather correlation information for esophageal contraction pattern recognition, and the generated CVP can automatically discover the vigor to determine the propagation in the scene.

4. Discussion

In contract, the esophageal manometry is one of the most traditional method used to assess esophageal motility patterns to measure the amplitude of contractile events in the esophagus and its sphincters in relation to time. The HRM represents an evolution from conventional esophageal manometry methods in which was incorporated up to 36 pressure sensors, and the data is converted into seamless and dynamic spatiotemporal esophageal pressure topography (EPT) by advanced software algorithms [4].

HRM is a diagnostic method in the diagnosis of specific esophageal motor disorders. It is considered as standard method for evaluation of esophageal motor function. However, due to the complexity of HRM data with many observational parameters, the doctors need to analyze a huge of data to obtain the diagnosis result which lead to time consuming and may also get diverse results from the same HRM image. Thus, to address this problem, we improved the efficiencies of the diagnosis of

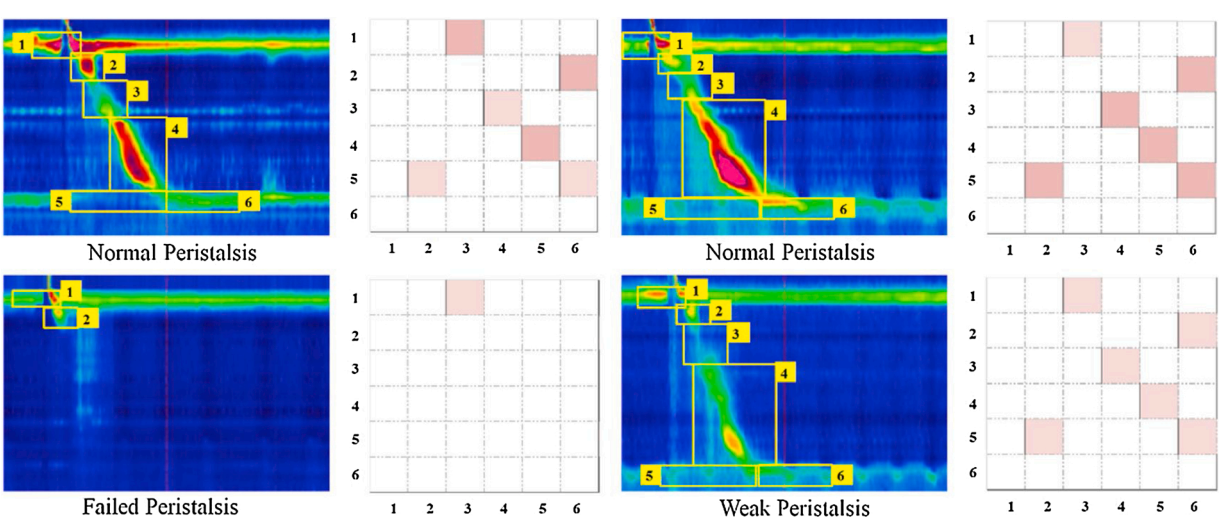

Fig. 3. Visualization of the learned propagation vigor correlation graph.
Each row illustrates two examples. For each example, we depict: the (1) input HRM image with group-truth bounding boxes and contractile vigor label; (2) matrix G of trained propagation correlation graph with ground-truth contractile vigor labels. In each HRM image, the vigor who with a high-pressure representation of G is denoted with in red; otherwise, it is depicted with light red.

esophageal motor disorders in clinical evaluation by using GCN on the HRM image to recognize the esophageal contraction pattern, i.e., classify the patients with esophageal symptoms in three classes such as normal, weak, and failed. In our experiment, the same data sets were used to compare our proposed method to existing approaches state-of-the-art deep learning. However, it is worth to mentioned that the size of our data sets is small which can be biased on accuracy.

Moreover, our experiment showed that the proposed method achieves 95.8 % in accuracy on the HRM data. We found that our accuracy surpasses slightly state-of-the-art deep learning method [47] with 0.2 %, while it outperformed others methods. This may be due to the small size of the HRM image dataset available and constituted the limitation of our study. Our study provided precision, Recall, F1-score (F1), and Coefficient of Determination (R2) of 96.5 %, 97.5 %, 95.2 %, and 93.5 %, respectively, while the traditional method such as Chicago Classification did not [13]. This can lead to the misclassification and false analysis in clinical evaluation. With a huge of the HRM data available, our method can achieve the best performance in accuracy and helps doctors to get the diagnosis of the esophageal motility disorders in the few minute.

5. Conclusions

This work presented an efficient and flexible method for determining the relevant correlation between propagation vigor in HRM scenes. We learn a contractile vigor propagation graph (CVP) and then perform graph correlational reasoning to recognize esophageal contraction patterns. We compared the proposed model with the state-of-the-art ConvNet models and achieved promising results. The comprehensive ablation studies demonstrate that the proposed model is able to learn the correlation information to understand contractile vigor propagation. In the future, we plan to further investigate how CVP works and to classify manometries based on the Chicago classification criteria to perform diagnoses.

CRediT authorship contribution statement

Zheng Wang: The idea the research, Methodology, Formal analysis, Writing - Original Draft and Editing. **Yuzhuo Dai:** Helping in programming. **Fanggen Lu:** Helping in the different steps of research. **Jie Zhang:** Helping in writing and editing of the paper. **Xiaowei Liu:** Software. **Muzhou Hou:** Supervision. **Lu Yan:** Review.

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Declaration of Competing Interest

The authors report no declarations of interest.

References

[1] G. Zambito, R. Roether, B. Kern, R. Conway, D. Scheeres, A. Banks-Venegoni, Is barium esophagram enough? Comparison of esophageal motility found on barium esophagram to high resolution manometry, Am. J. Surg. (2020).
[2] A. Sawada, M. Guzman, K. Nikaki, S. Sonmez, E. Yazaki, Q. Aziz, P. Woodland, B. Rogers, C.P. Gyawali, D. Sifrim, Identification of different phenotypes of esophageal reflux hyper-sensitivity and implications for treatment, Clin. Gastroenterol. Hepatol. (2020).
[3] M.Y. Zhang, N.D. Tan, Y.W. Li, D. Sifrim, J.E. Pandolfino, Y.L. Xiao, M.H. Chen, Esophageal symptoms versus epigastric symptoms: relevance for diagnosis of gastroesophageal reflux disease, J. Dig. Dis. 21 (12) (2020) 696–704.
[4] K. Kunieda, I. Fujishima, H. Wakabayashi, T. Ohno, T. Shigematsu, M. Itoda, F. Oshima, T. Mori, N. Ogawa, S. Ogawa, Relationship between tongue pressure and pharyngeal function assessed using high-resolution manometry in older dysphagia patients with sarcopenia: a pilot study, Dysphagia 36 (1) (2021) 33–40.
[5] L. Peng, A. Patel, V. Kushnir, C.P. Gyawali, Assessment of upper esophageal sphincter function on high-resolution manometry: identification of predictors of globus symptoms, J. Clin. Gastroenterol. 49 (2) (2015) 95.
[6] J.E. Pandolfino, S.K. Ghosh, Q. Zhang, A. Jarosz, N. Shah, P.J. Kahrilas, Quantifying egj morphology and relaxation with high-resolution manometry: a study of 75 asymptomatic volunteers, Am. J. Physiol. Gastrointestinal Liver Physiol. 290 (5) (2006) G1033–G1040.
[7] P.J. Kahrilas, S.K. Ghosh, J.E. Pandolfino, Esophageal motility disorders in terms of pressure topography: the chicago classification, J. Clin. Gastroenterol. 42 (5) (2008) 627.
[8] A. Rengarajan, C.P. Gyawali, High-resolution manometry can characterize esophagogastric junction morphology and predict esophageal reflux burden, J. Clin. Gastroenterol. 54 (1) (2020) 22–27.
[9] H. Taniguchi, Y. Aoyagi, K. Matsuo, S. Imaeda, M. Hirumuta, E. Saitoh, Elicitation of the swallowing reflex by esophageal stimulation in healthy subjects: an evaluation using high-resolution manometry, Dysphagia 35 (4) (2020) 657–666.
[10] S.K. Ghosh, J.E. Pandolfino, J. Rice, J.O. Clarke, M. Kwiatek, P.J. Kahrilas, Impaired deglutitive egj relaxation in clinical esophageal manometry: a quantitative analysis of 400 patients and 75 controls, Am. J. Physiol. Gastrointestinal Liver Physiol. 293 (4) (2007) G878–G885.
[11] F.B. Van Hoeij, A.J. Bredenoord, Clinical application of esophageal high-resolution manometry in the diagnosis of esophageal motility disorders, J. Neurogastroenterol. Motil. 22 (1) (2016) 6.
[12] K. Aoyama, K. Kunieda, T. Shigematsu, T. Ohno, I. Fujishima, Effect of bridge position swallow on esophageal motility in healthy individuals using high-resolution manometry, Dysphagia (2020) 1–7.
[13] H. Foisy, M. Pioche, E. Chabrun, T. Ponchon, F. Zerbib, J. Rivory, F. Mion, S. Roman, Role of rapid drink challenge during esophageal high-resolution manometry in predicting outcome of peroral endoscopic myotomy in patients with achalasia, J. Neurogastroenterol. Motil. 26 (2) (2020) 204.
[14] L.G. Alcala Gonzalez, R.A. Oude Nijhuis, A.J. Smout, A.J. Bredenoord, Normative reference values for esophageal high-resolution manometry in healthy adults: a systematic review, Neurogastroenterol. Motil. 33 (1) (2021), e13954.
[15] E. Soffer, M. Pimentel, J. Conklin, Color Atlas of High Resolution Manometry, Springer US, 2009.
[16] D. Carlson, P. Kahrilas, How to effectively use high-resolution esophageal manometry, Gastroenterology 151 (5) (2016) 789–792.
[17] W. Kou, D.A. Carlson, A.J. Baumann, E. Donnan, Y. Luo, J.E. Pandolfino, M. Etemadi, A deep-learning-based unsupervised model on esophageal manometry using variational autoencoder, Artif. Intell. Med. (2021), 102006.
[18] D.A. Carlson, W. Kou, K.P. Rooney, A.J. Baumann, E. Donnan, J.R. Triggs, E. N. Teitelbaum, A. Holmstrom, E. Hungness, S. Sethi, et al., Achalasia subtypes can be identified with functional luminal imaging probe (flip) panometry using a supervised machine learning process, Neurogastroenterol. Motil. (2020), e13932.
[19] E. Hewson, D. Ott, C. Dalton, Y. Chen, W. Wu, J. Richter, Manometry and radiology. Complementary studies in the assessment of esophageal motility disorders, Gastroenterology 98 (1990) 626–632.
[20] C.P. Gyawali, Optimizing the high-resolution manometry (hrm) study protocol, Neurogastroenterol. Motil. 27 (2) (2021).
[21] F. van Hoeij, A. Bredenoord, Clinical application of esophageal high-resolution manometry in the diagnosis of esophageal motility disorders, J. Neurogastroenterol. Motil. 22 (2021).
[22] R. Yadlapati, High-resolution esophageal manometry: interpretation in clinical practice, Curr. Opin. Gastroenterol. 33 (2017) 1.
[23] W.O.A. Rohof, A.J. Bredenoord, Chicago classification of esophageal motility disorders: lessons learned, Curr. Gastroenterol. Rep. 19 (8) (2017) 37.
[24] P. Kahrilas, A. Bredenoord, M. Fox, C. Gyawali, S. Roman, A. Smout, J. Pandolfino, The chicago classification of esophageal motility disorders, v3.0, Neurogastroenterol. Motil. 27 (2021).
[25] T. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, 2021.
[26] S. Miller, M. Ptok, M. Jungheim, Influence of acid swallows on the dynamics of the upper esophageal sphincter, Dysphagia (2020) 1–13.
[27] J. Bruna, W. Zaremba, A. Szlam, Y. Lecun, Spectral Networks and Locally Connected Networks on Graphs, 2021.
[28] M. Defferrard, X. Bresson, P. Vandergheynst, Convolutional neural networks on graphs with fast localized spectral filtering, in: Proceedings of the 30th International Conference on Neural Information Processing Systems, NIPS16, 2016, 3844C3852.
[29] M. Niepert, M. Ahmed, K. Kutzkov, Learning Convolutional Neural Networks for Graphs, 2021.
[30] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Lio, Y. Bengio, Graph Attention Networks, 2021.
[31] H. Kaiming, G. Georgia, D. Piotr, G. Ross, Mask r-cnn, IEEE Trans. Pattern Anal. Mach. Intell. (2017), 1–1.
[32] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: towards real-time object detection with region proposal networks, IEEE Trans. Pattern Anal. Mach. Intell. 39 (6) (2016) 1137–1149.
[33] A. Santoro, D. Raposo, D. Barrett, M. Malinowski, R. Pascanu, P. Battaglia, T. Lillicrap, A simple neural network module for relational reasoning, NIPS17 (2017) 4974–4983.
[34] X. Wang, H. Ji, C. Shi, B. Wang, Y. Ye, P. Cui, P.S. Yu, Heterogeneous graph attention network, The World Wide Web Conference (2019) 2022–2032.

[35] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, NIPS17 (2017) 6000–6010.
[36] D. Bahdanau, K. Cho, Y. Bengio, Neural machine translation by jointly learning to align and translate, International Conference on Learning Representations (ICLR) (2015).
[37] J. Xiao, H. Ye, X. He, H. Zhang, F. Wu, T.S. Chua, Attentional Factorization Machines: Learning the Weight of Feature Interactions Via Attention Networks, 2021.
[38] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, Y. Bengio, Show, attend and tell: neural image caption generation with visual attention, Comput. Sci. (2015) 2048–2057.
[39] N. Pappas, A. Popescu-Belis, Multilingual Hierarchical Attention Networks for Document Classification, 2017.
[40] M.T. Luong, H. Pham, C.D. Manning, Effective approaches to attention-based neural machine translation, Comput. Sci. (2021).
[41] B.D. Rogers, A. Rengarajan, A. Mauro, M. Ghisa, N. De Bortoli, M. Cicala, M. Ribolsi, R. Penagini, E. Savarino, C.P. Gyawali, Fragmented and failed swallows on esophageal high-resolution manometry associate with abnormal reflux burden better than weak swallows, Neurogastroenterol. Motil. 32 (2) (2020), e13736.
[42] T. Fawcett, An introduction to roc analysis, Pattern Recognit. Lett. 27 (8) (2006) 861–874.
[43] Z. Wang, Y. Xiao, F. Weng, X. Li, Y. Meng, R-jaunlab: automatic multi-class recognition of jaundice on photos of subjects with region annotation networks, J. Digit. Imaging 9 (2021).
[44] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (2018) 8759–8768.
[45] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the inception architecture for computer vision, 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR) (2016) 2818–2826.
[46] K. Simonyan, A. Zisserman, Very Deep Convolutional Networks for Large-scale Image Recognition, 2021.
[47] L. Huang, P. Chang, J. Conklin, Tu1256cdeep learning for esophageal manometry? A proof of concept, Gastroenterology 156 (2019). S–1006.
[48] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (2018) 8759–8768.
[49] E. Xie, P. Sun, X. Song, W. Wang, P. Luo, Polarmask: single shot instance segmentation with polar representation, 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2020).